

CachePool: Many-core cluster of customizable, lightweight scalar-vector PEs for irregular L2 data-plane workloads

Integrated Systems Laboratory (ETH Zürich)

Zexin Fu, Diyou Shen

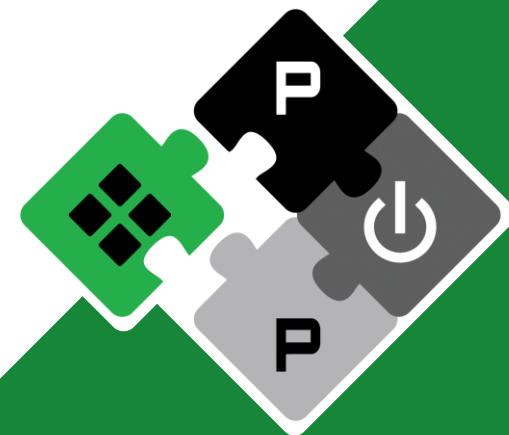
zexifu, dishen@iis.ee.ethz.ch

Alessandro Vanelli-Coralli
Luca Benini

avanelli@iis.ee.ethz.ch
lbenini@iis.ee.ethz.ch

PULP Platform

Open Source Hardware, the way it should be!



@pulp_platform



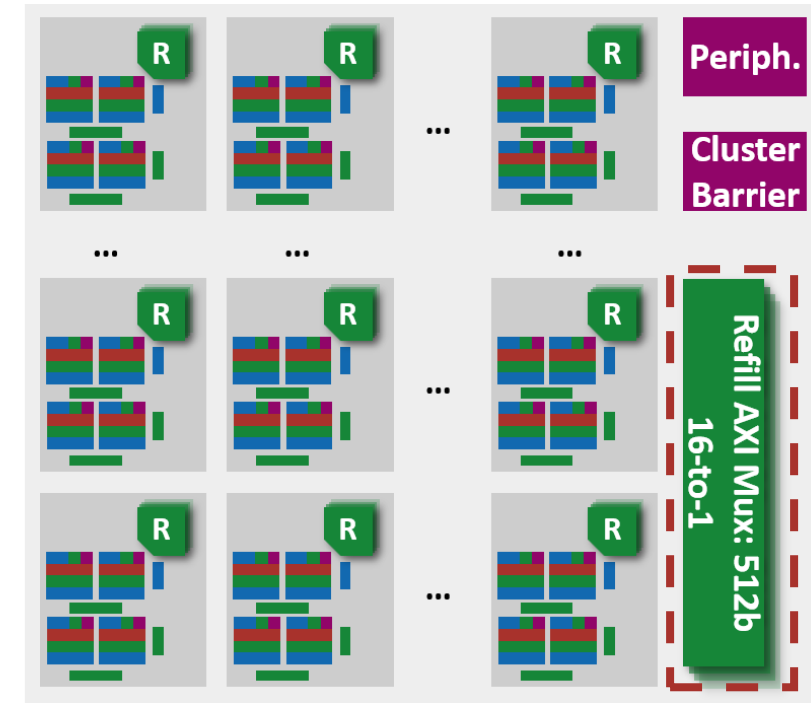
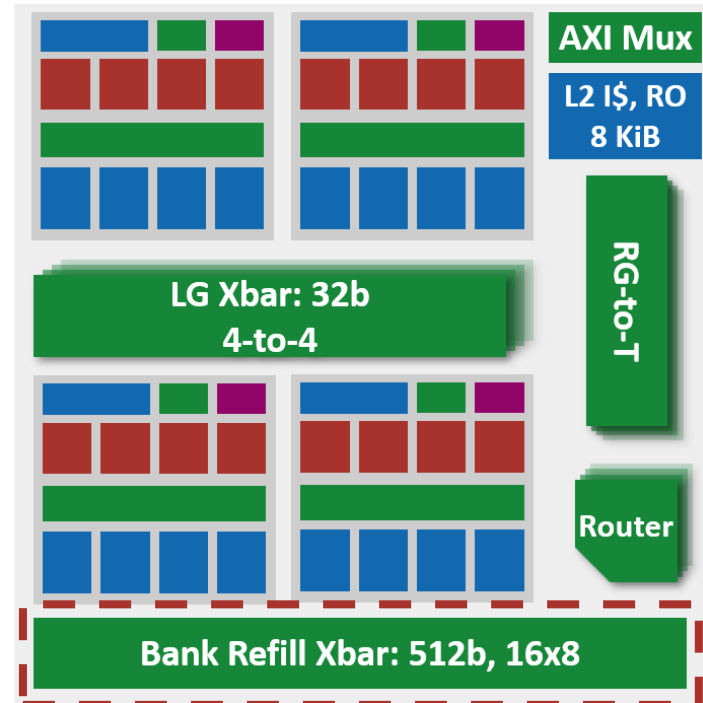
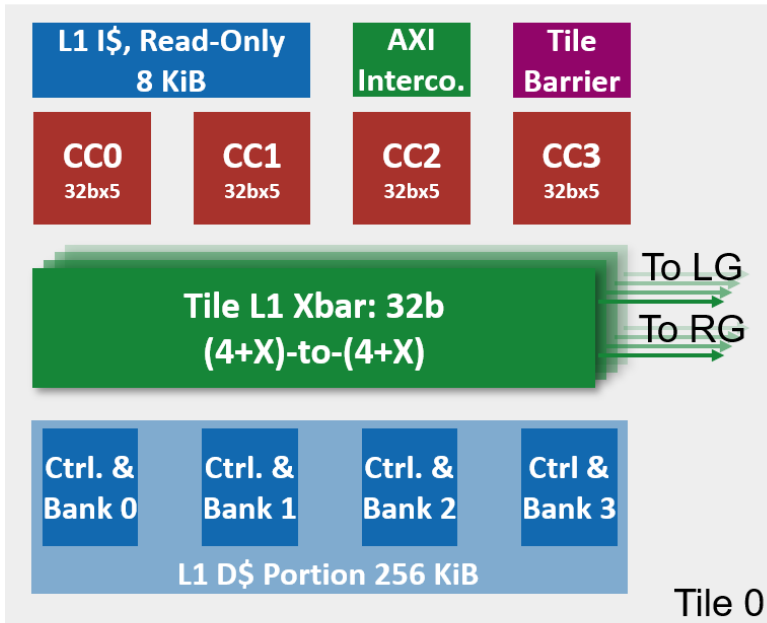
pulp-platform.org



youtube.com/pulp_platform



Hardware Development (Cluster)



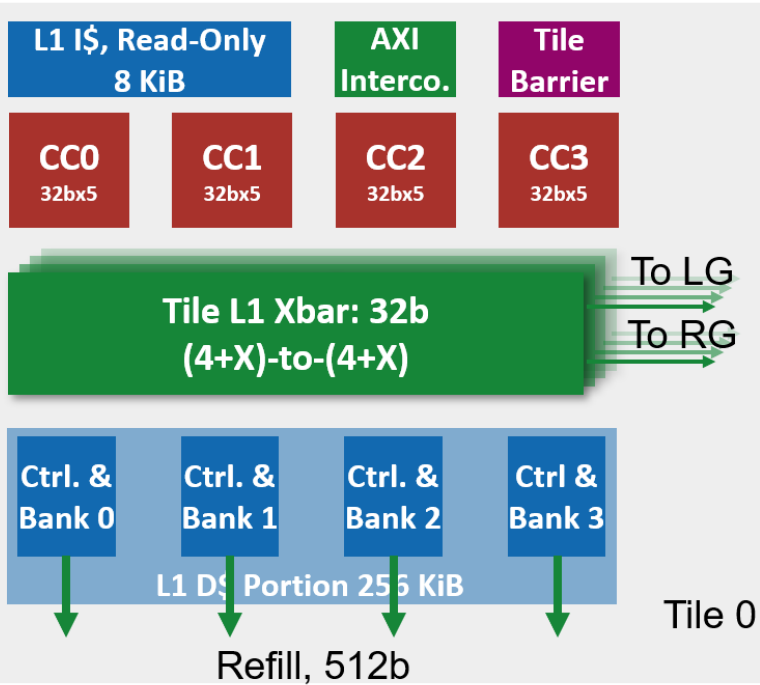
WIP: change to NoC



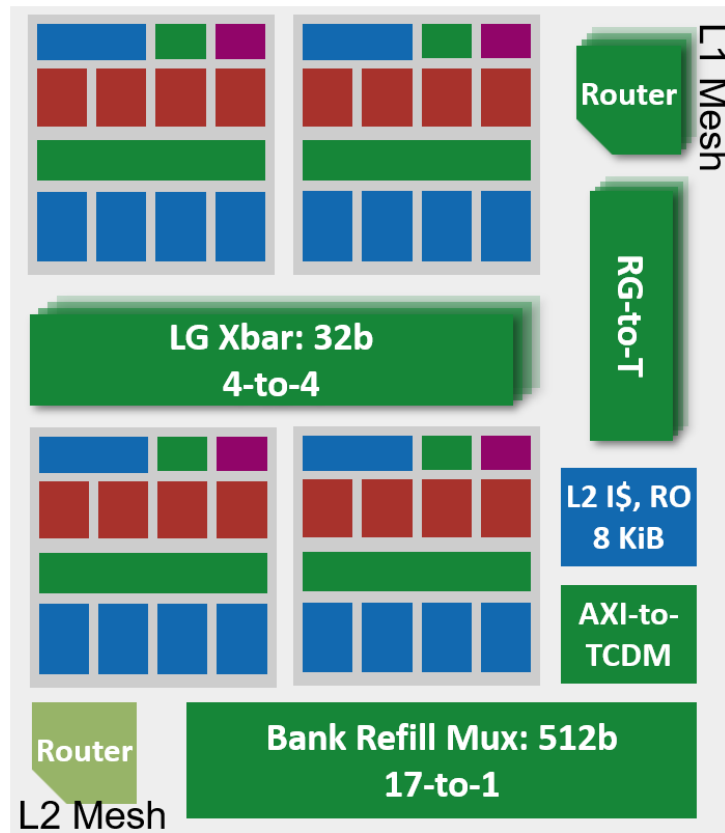
Hardware Development (Cluster)



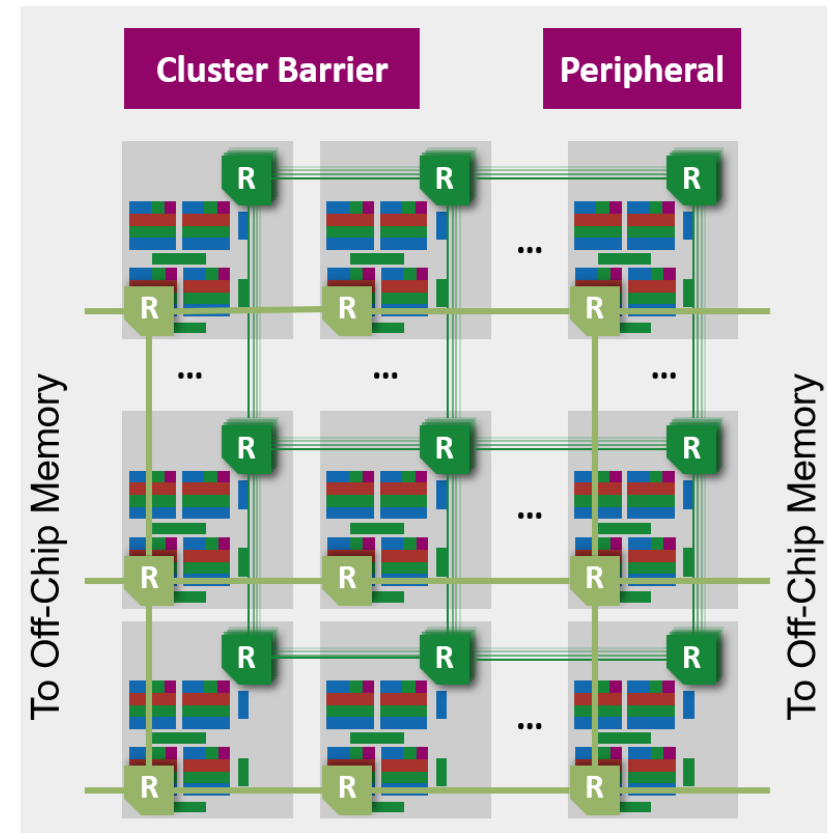
Tile



Group



Cluster



Hardware Development (Cluster)



- **Updated Block Diagram**
- **Refill NoC**
 - Add the L2 Refill NoC into the cluster
 - Currently implement using AXI NoC based on FlooNoC
 - Similar performance and area issue due to limitations in our AXI modules
AXI designed for long bursts (DMA transfers)
Cache refills are multiple 512b small transactions => requires lots of AXI IDs (area, timing)
 - WIP to switch to TCDM NoC
Similar to L1 NoC

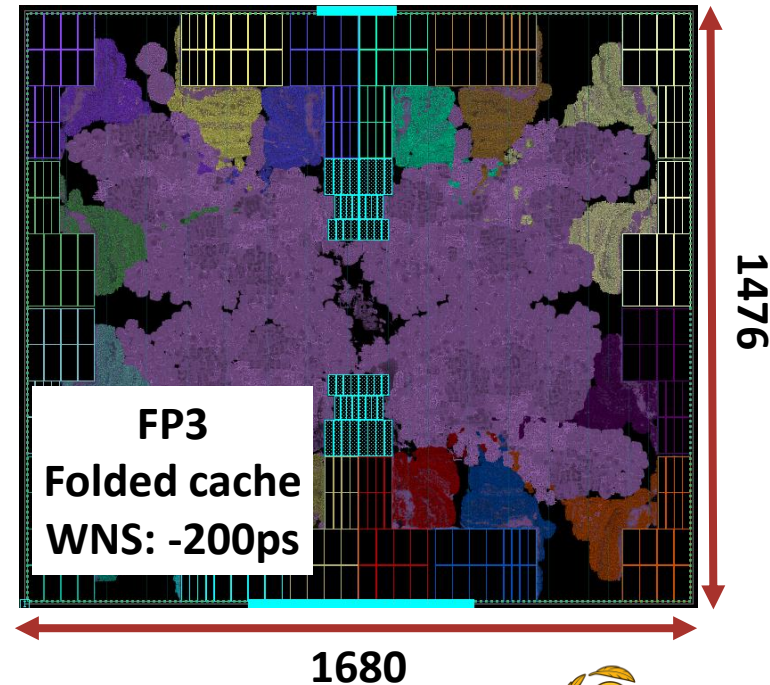
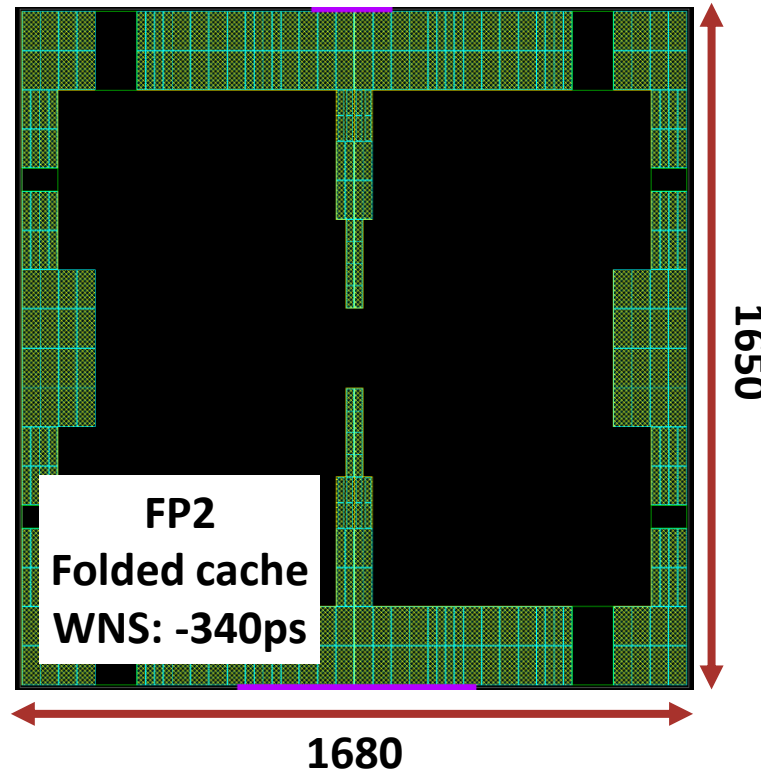
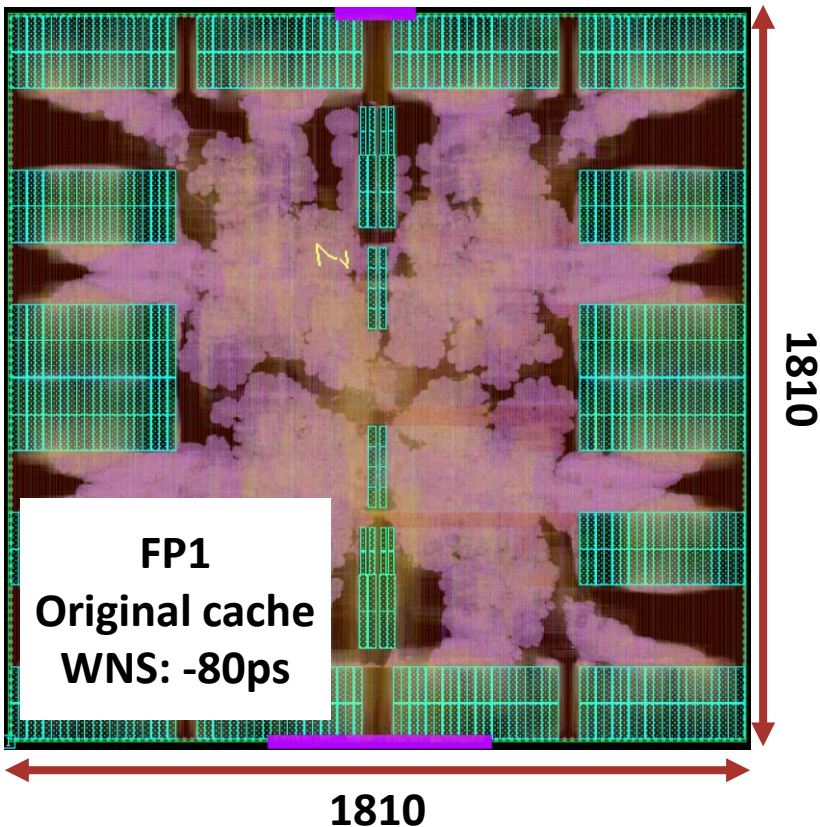


Physical Design (Folded Cache)

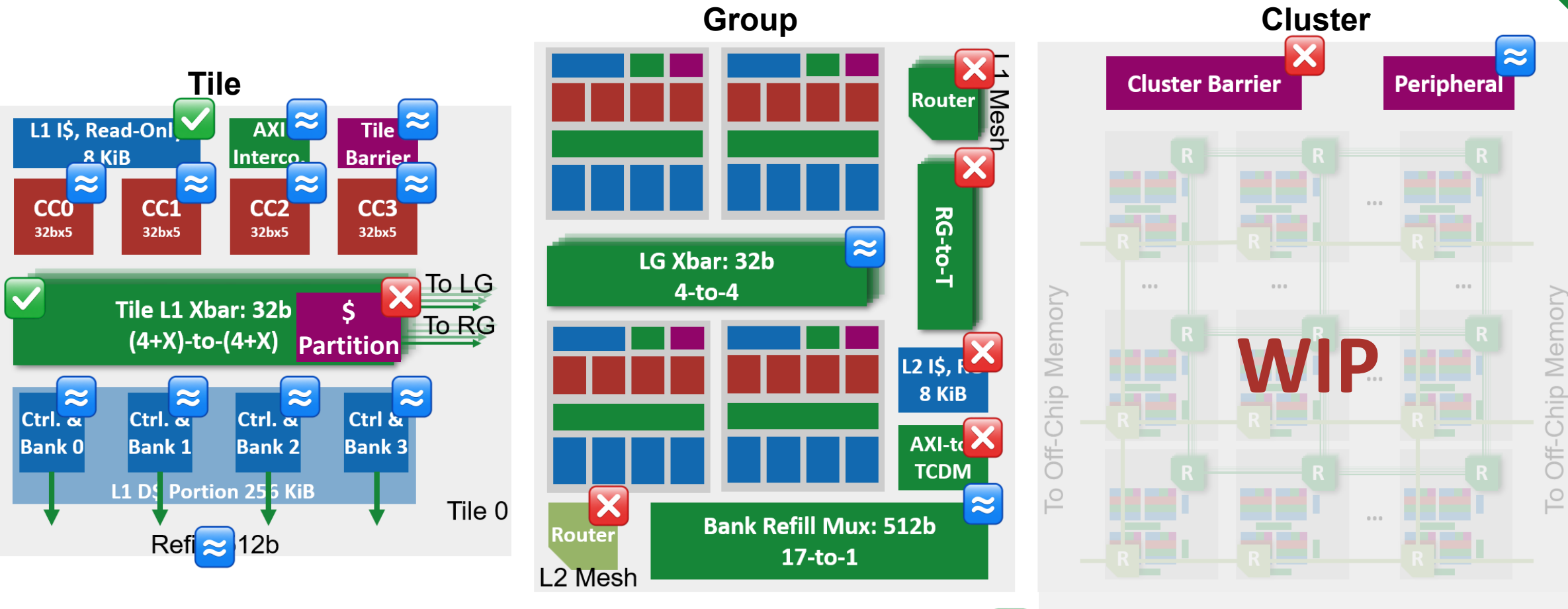


- **Timing**

- ~300ps WNS in FP2; Path: Tag -> controller -> Tag
- ~200ps WNS in FP3. Zexin worked on RTL-side timing improvements (also reduced to ~220ps)



GVSoc Modeling Progress



- ✓ Done, the same logic as RTL
- ≈ In place, with approximate model
- ✗ Not implemented yet

GVSoC Modeling Progress



- **Core-Level Modeling**

- Zexin has added and calibrated some missing supports in the cache model

- **Cluster-Level Modeling**

- Zexin has built the tile-level model, and further scale it to multi-tile
- Data corruption encountered when working using multiple cores
 - Still debugging
- Further scaling is WIP, and has the highest priority on the task list



Next Step

- **Diyou Side**

- Mainly focus on GVSoC modeling now
- Improve refill NoC
- Timing improvement for folded cache



Timeline



		2025							2026					
		11	12	1	2	3	4	5	6	7	8	9	10	11
WP1 ManyCore Architecture	Multi-Tile Scaling [RTL] Complete													
	4/16-Group Scaling [GVSoC + RTL] Ongoing													
	Interco. Optimization (multi-group) [RTL] Ongoing													
	Timing optimization for NoC [RTL] Ongoing by colleague													
	NoC Topology Exploration [RTL, GVSoC] Ongoing by student													
WP2 Cache	Cache partitioning [RTL] Complete													
	WT-L1 by master student [RTL] Complete, Extending Ongoing													
	Folded Cache [GVSoC + RTL] Complete, Opt Ongoing													

Timeline



		2025								2026				
		11	12	1	2	3	4	5	6	7	8	9	10	11
WP3 VPE	Shared vector core [RTL + GVSoc] Ongoing													
WP4 Benchmark	Single-user kernel optimization [SW] Ongoing													
	Multi-user and more scenarios [SW]													
Report	Final Report and Benchmark													

L1 Access Latency



- **Problem: increased L1 access latency when scaling up**
- **Solutions**
 - **(Ongoing, RTL & Physical) Hardware optimization**
Further optimize the interconnection and cache controller to reduce access latency on existing architecture
=> Retiming the current interconnection, remove unnecessary cuts in interco and cache
=> Target to reduce 1-2 cycles (~5 cyc) for hit-to-use latency
Currently has reduced to 6 cycles:
Core => Xbar => AMO => Ctrl (2) => AMO => Core
 - **(Complete, RTL) Cache partitioning**
Partition the shared L1 to keep data localized to avoid remote access latency
 - **(Complete with some problems, RTL & Physical) WT-L1**
Explore and evaluate a private WT L1 for each core
=> Performance degrade on single tile
=> Needs more work to support larger system



L1 Access Latency



- **Problem: large NoC latency (~28 cyc based on previous exploration)**
- **Solutions**
 - **(Ongoing, Physical) Reduce hop latency**
Explore the timing under 7nm technology to see the possibility of reducing per-hop latency to 1 cycle
=> Reduce the latency by 1/2
 - **(Planned, Physical) Cut clk-domain (optional)**
If 1-cyc is not possible, explore to operate the NoC under a higher frequency than cores (e.g. 1.5GHz)
=> Effectively cut the latency by 1/3
 - **(Planned, GVSoc & Physical) NoC Topology**
Explore different NoC topologies
=> Torus to reduce the diameter of NoC by 1/2, reduce longest latency by 1/2



PE-Related



- **Problem: Improve scalar core performance**
- **Solutions**
 - **(Planned, RTL & GVSoc) Shared vector core**
Explore two Snitch sharing one Spatz configuration to improve scalar performance
Software complexity needs to be considered, especially on the possible conflicting use of register file.
Discussed Solution: critical section



RLC Workload



- **Problem: Current RLC model cannot reveal all properties of the kernel**
- **Solutions**
 - **(Ongoing, SW) Kernel optimization**
Adapt the current model to better represent the real cases, e.g. do some operations on the data. Further cooperation with HQ side to develop a more complete and better model.
 - **(Planned, SW) Kernel development**
Implement the missing use cases for multi-user



Thank you!

Q&A

